Design Database

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Original Flat Relation

The original flat relation is derived by combining all columns from the CSV file (visit table) and the SQLite database tables (restaurants and servers):

R(RestaurantID, RestaurantName, City, State, hasTableService, EmpID, ServerName, StartDateHired, EndDateHired, HourlyRate, BirthDate, SSN, VisitDate, VisitTime, MealType, PartySize, WaitTime, CustomerName, CustomerPhone, CustomerEmail, LoyaltyMember, FoodBill, TipAmount, FoodDiscountPercentage, PaymentMethod, OrderedAlcohol, AlcoholBill)

Restaurant Table

Primary Key for this could be `rID`

- $rID \rightarrow RestaurantName$
- $rID \rightarrow city$
- $rID \rightarrow hasTableService$
- $city \rightarrow state$

This relation violates 3NF due to the following transitive dependencies: - $city \rightarrow state$ (city determines state, but city is not a key attribute)

Servers Table

Primary Key for this could be `EmpID`

- $EmpID \rightarrow ServerName$
- $EmpID \rightarrow StartDateHired$
- $EmpID \rightarrow EndDateHired$
- $EmpID \rightarrow HourlyRate$
- $EmpID \rightarrow BirthDate$
- $EmpID \rightarrow ssn$
- $EmpID \rightarrow RestaurantID$

CSV File Table (Visit Table)

Primary Key for this table should be a added attribute called `visitID`

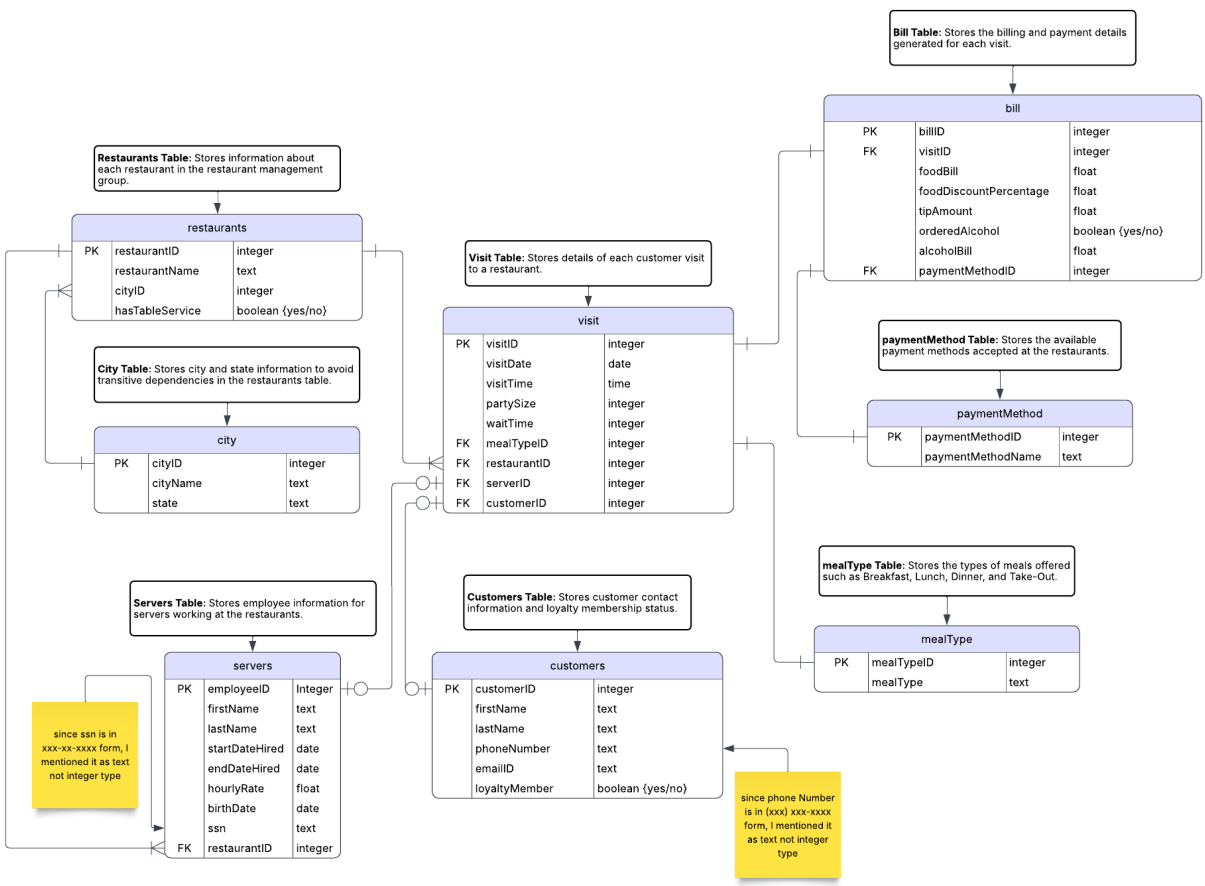
- $ServerEmpID \rightarrow ServerName$
- $ServerEmpID \rightarrow RestaurantID$
- $CustomerEmail \rightarrow LoyaltyMember$
- $CustomerPhone \rightarrow LoyaltyMember$

All these dependencies violate 3NF because non-key attributes determine other non-key attributes, causing transitive dependencies. These are resolved by decomposing into 8 normalized tables.

- `MealType` repeating values → decomposed into `mealType table`
- `PaymentMethod` repeating values → decomposed into `paymentMethod table`

I have created 8 tables to combine the 2 tables (servers and restaurants) from sqlite database and the 1 table from the cvs file. All the tables are in 3NF form.

ER Diagram



Functional Dependencies and Normalization

1. Servers Table

Functional Dependencies:

- $employeeID \rightarrow firstName$
- $employeeID \rightarrow lastName$
- $employeeID \rightarrow startDateHired$
- $employeeID \rightarrow endDateHired$
- $employeeID \rightarrow hourlyRate$
- $employeeID \rightarrow birthDate$
- $employeeID \rightarrow ssn$
- $employeeID \rightarrow restaurantiD$

Explanation of table being in 3NF:

- 1NF: All attributes are atomic and single-valued. There are no repeating groups.
- 2NF: The primary key is a single attribute `employeeID`, therefore partial dependencies are not possible and the table is in 2NF.
- 3NF: No non-key attribute determines another non-key attribute. All attributes `firstName`, `lastName`,`startDateHired`, `endDateHired`, `hourlyRate`, `birthDate`, `ssn`, `restaurantID` depend directly and solely on `employeeID`. Therefore no transitive dependencies exist.

2. Restaurants Table

Functional Dependencies:

- $restaurantID \rightarrow restaurantName$
- $restaurantID \rightarrow citylD$
- $restaurantID \rightarrow hasTableService$

Explanation of table being in 3NF:

- 1NF: All attributes are atomic and single-valued. There are no repeating groups.
- 2NF: The primary key is a single attribute `restaurantID`, therefore partial dependencies are not possible and the table is in 2NF.
- 3NF: No non-key attribute determines another non-key attribute. All attributes `restaurantName`, `citylD`,`hasTableService` depend directly and solely on `restaurantID`. Therefore no transitive dependencies exist.

3. City Table

Functional Dependencies:

- $cityID \rightarrow cityName$
- $cityID \rightarrow state$

Note: This table is a lookup/associate table to decompose the transitive relationship that would have occurred in the **Restaurants Table** if we kept `city` and `state` both in it. As $city \rightarrow state$ where city is not a key attribute.

Explanation of table being in 3NF:

- 1NF: All attributes are atomic and single-valued. There are no repeating groups.
- 2NF: The primary key is a single attribute `cityID`, therefore partial dependencies are not possible and the table is in 2NF.
- 3NF: No non-key attribute determines another non-key attribute. The only other attribute `cityName` depends directly and solely on `cityID`. Therefore no transitive dependencies exist.

4. Visit Table

Functional Dependencies:

- $visitID \rightarrow visitDate$
- $visitID \rightarrow visitTime$
- $visitID \rightarrow partySize$
- $visitID \rightarrow waitTime$
- $visitID \rightarrow mealTypelD$
- $visitID \rightarrow restaurantiD$
- $visitID \rightarrow serverlD$
- $visitID \rightarrow customerID$

Explanation of table being in 3NF:

- 1NF: All attributes are atomic and single-valued. There are no repeating groups.
- 2NF: The primary key is a single attribute `visitID`, therefore partial dependencies are not possible and the table is in 2NF.
- 3NF: No non-key attribute determines another non-key attribute. All attributes `visitDate`, `visitTime`, `partySize`, `waitTime`, `mealTypelD`, `restaurantiD`, `serverlD`, `customerID` depend directly and solely on `visitID`. Therefore no transitive dependencies exist.

5. Customers Table

Functional Dependencies:

- $customeriD \rightarrow firstName$
- $customeriD \rightarrow lastName$
- $customeriD \rightarrow phoneNumber$
- $customeriD \rightarrow emaillD$
- $customeriD \rightarrow loyaltyMember$

Explanation of table being in 3NF:

- 1NF: All attributes are atomic and single-valued. There are no repeating groups.
- 2NF: The primary key is a single attribute `customeriD`, therefore partial dependencies are not possible and the table is in 2NF.
- 3NF: No non-key attribute determines another non-key attribute. All attributes `firstName`, `lastName`, `phoneNumber`, `emaillD`, `loyaltyMember` depend directly and solely on `customeriD`. Therefore no transitive dependencies exist.

6. Bill Table

Functional Dependencies:

- $billiD \rightarrow visitiD$
- $billiD \rightarrow foodBill$
- $billiD \rightarrow foodDiscountPercentage$
- $billiD \rightarrow tipAmount$
- $billiD \rightarrow orderedAlcohol$
- $billiD \rightarrow alcoholBill$
- $billiD \rightarrow paymentMethodID$

Explanation of table being in 3NF:

- 1NF: All attributes are atomic and single-valued. There are no repeating groups.
- 2NF: The primary key is a single attribute `billiD`, therefore partial dependencies are not possible and the table is in 2NF.
- 3NF: No non-key attribute determines another non-key attribute. All attributes `visitiD`, `foodBill`, `foodDiscountPercentage`, `tipAmount`, `orderedAlcohol`, `alcoholBill`, `paymentMethodID` depend directly and solely on `billiD`. Therefore no transitive dependencies exist.

7. Payment Method Table

Functional Dependencies:

- $paymentMethodiD \rightarrow paymentMethodName$

Note: This table is a lookup/associate table to decompose the multiple-choice value that could have been in the payment option such as `credit card`, `mobile payment`, `cash` etc. This ensures that if a new payment method is added then it only needs to be updated here instead of changing the `check constraint` in the **bill table** for the `paymentMethod` attribute.

Explanation of table being in 3NF:

- 1NF: All attributes are atomic and single-valued. There are no repeating groups.
- 2NF: The primary key is a single attribute `paymentMethodiD`, therefore partial dependencies are not possible and the table is in 2NF.
- 3NF: No non-key attribute determines another non-key attribute. The only other attribute `paymentMethodName` depends directly and solely on `paymentMethodiD`. Therefore no transitive dependencies exist.

8. Meal Type Table

Functional Dependencies:

- $mealTypelD \rightarrow mealType$

Explanation of table being in 3NF:

- 1NF: All attributes are atomic and single-valued. There are no repeating groups.
- 2NF: The primary key is a single attribute `mealTypelD`, therefore partial dependencies are not possible and the table is in 2NF.
- 3NF: No non-key attribute determines another non-key attribute. The only other attribute `mealType` depends directly and solely on `mealTypelD`. Therefore no transitive dependencies exist.